Digital Decimation Filter: CIC + FIR Droop Compensator

Inter-IIT VLSI Bootcamp, IIT Guwahati

GitHub repository (Verilog sources): PASTE-YOUR-GITHUB-LINK-HERE

1. Architecture

Signal flow (20-bit output, effective rates after the 2048 decimation):

2-bit input bitstream > CIC (3 integrators, down-sample by 2048, 3 combs) > drop 10 LSBs > 3-tap symmetric FIR at the output rate > round to 20 bits

Block	Choice / reason
CIC order N=3, R=2048, M=1	Rejects out-of-band modulator noise with adders only. R absorbs the whole OSR of 2048.
Register width 35 bits	Hogenauer bound: N*log2(R*M)+Bin = 3*11+2 = 35. Integrators wrap in two's complement; combs cancel the wrap, so no clipping inside the CIC.
Task 1 output	Round and drop 13 LSBs to get 20 bits (largest scale that does not clip).
FIR input	CIC output >> 10 (25 bits), so the FIR keeps extra precision before the final 20-bit rounding.
FIR compensator (Task 2)	3-tap symmetric, runs at the decimated rate. Gain is folded into the coefficients so the largest tone uses ~98% of the 20-bit range. Final output rounded to 20 bits.

2. Coefficients and justification

Tap	Integer value	Real value (/2^22)	Note
h[0] = h[2]	-530538	-0.1265	outer taps (symmetric)
h[1]	1768461	0.4216	centre tap
Format	22-bit signed	shift right 22	output rounded (+2^21)

The CIC response is sinc^3, which droops by about 11.8 dB at 990 Hz. A symmetric 3-tap FIR has response H(f) = h1 + 2*h0*cos(2*pi*f/Fs). With h0/h1 = -0.3 this is unity-shaped at DC and rises toward Nyquist, giving about +12.0 dB at 990 Hz, which nearly cancels the droop (cascade about +0.3 dB at 990 Hz). The ratio -0.3 was chosen by a sweep that maximised the worst-case SNR across the 10, 500 and 990 Hz tones after 20-bit quantisation. Longer filters were tested and did not improve SNR, because the 20-bit rounding, not the droop shape, becomes the limit.

SNR RESULTS

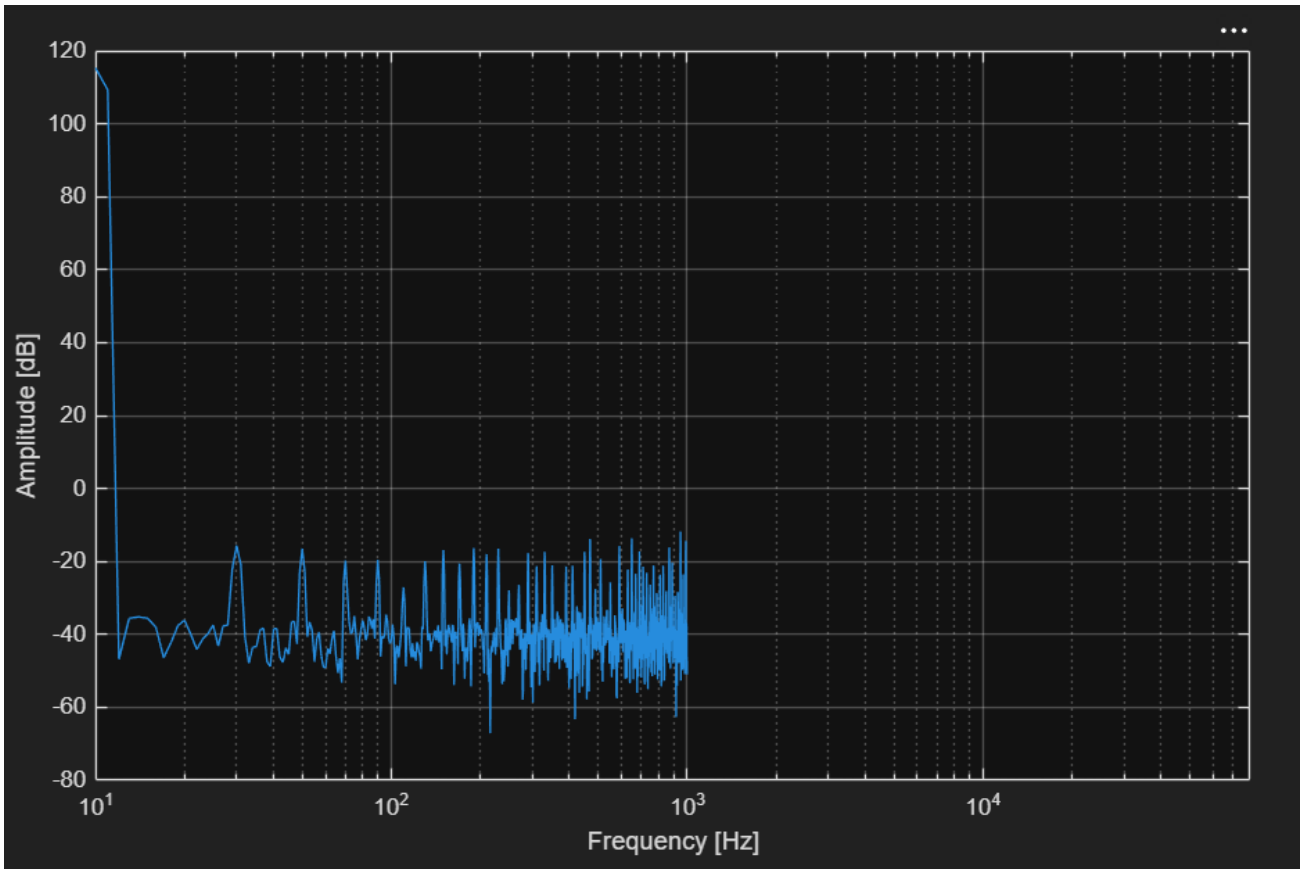
Tone	Task 1: CIC only (dB)	Task 2: CIC + FIR (dB)	Change (dB)
10 Hz	117.11	118.84	+1.7
500 Hz	Inf* (about 159.3)	123.76	n/a
990 Hz	105.86	119.31	+13.5

* At 500 Hz the Task 1 output is an exactly repeating 4-sample pattern, so the noise power falls below the floating-point resolution of the script (total minus signal) and it returns Inf. Summing the noise bins directly gives about 159.3 dB. This is a measurement artefact of a coherent test tone, not a real 159 dB capability. The 10 Hz and 990 Hz values are the representative ones. The largest Task 2 output peak is 516424, below the 20-bit limit of 524287 (no clipping).

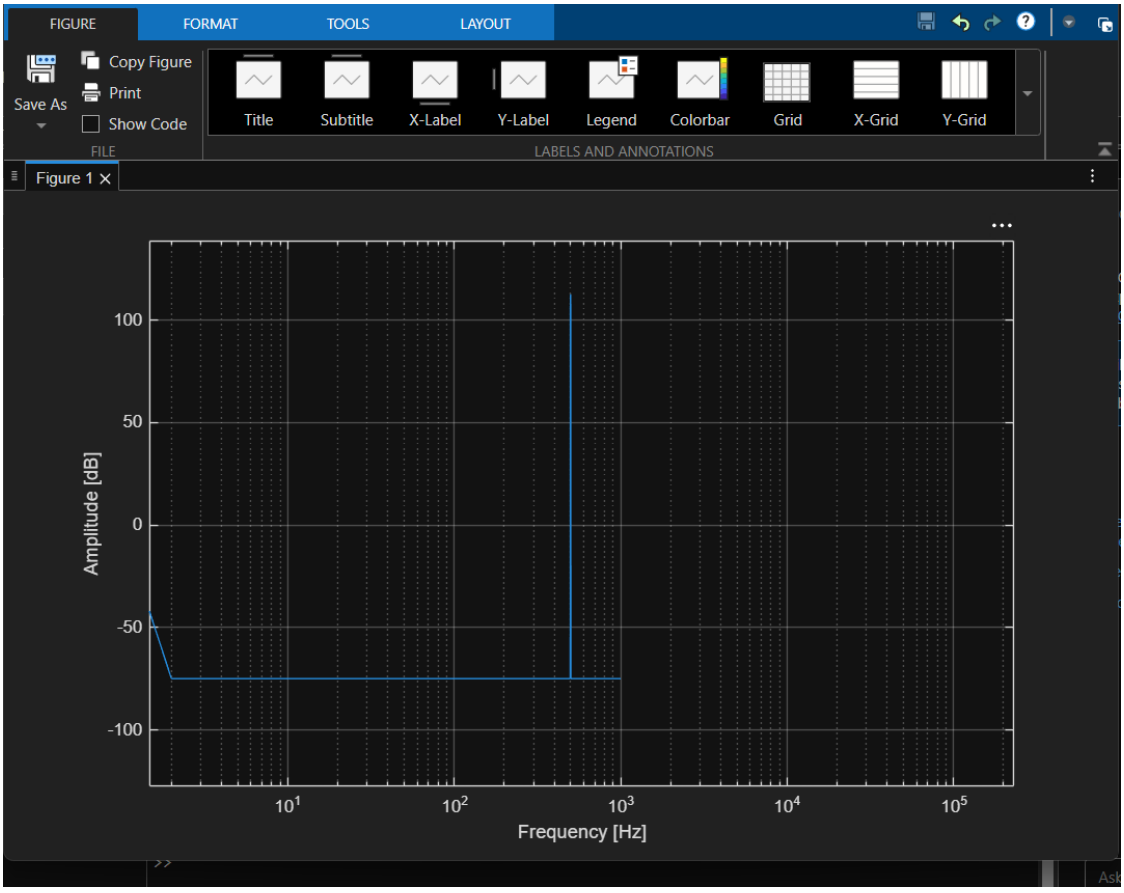
3. PSD plots

1. TASK 1

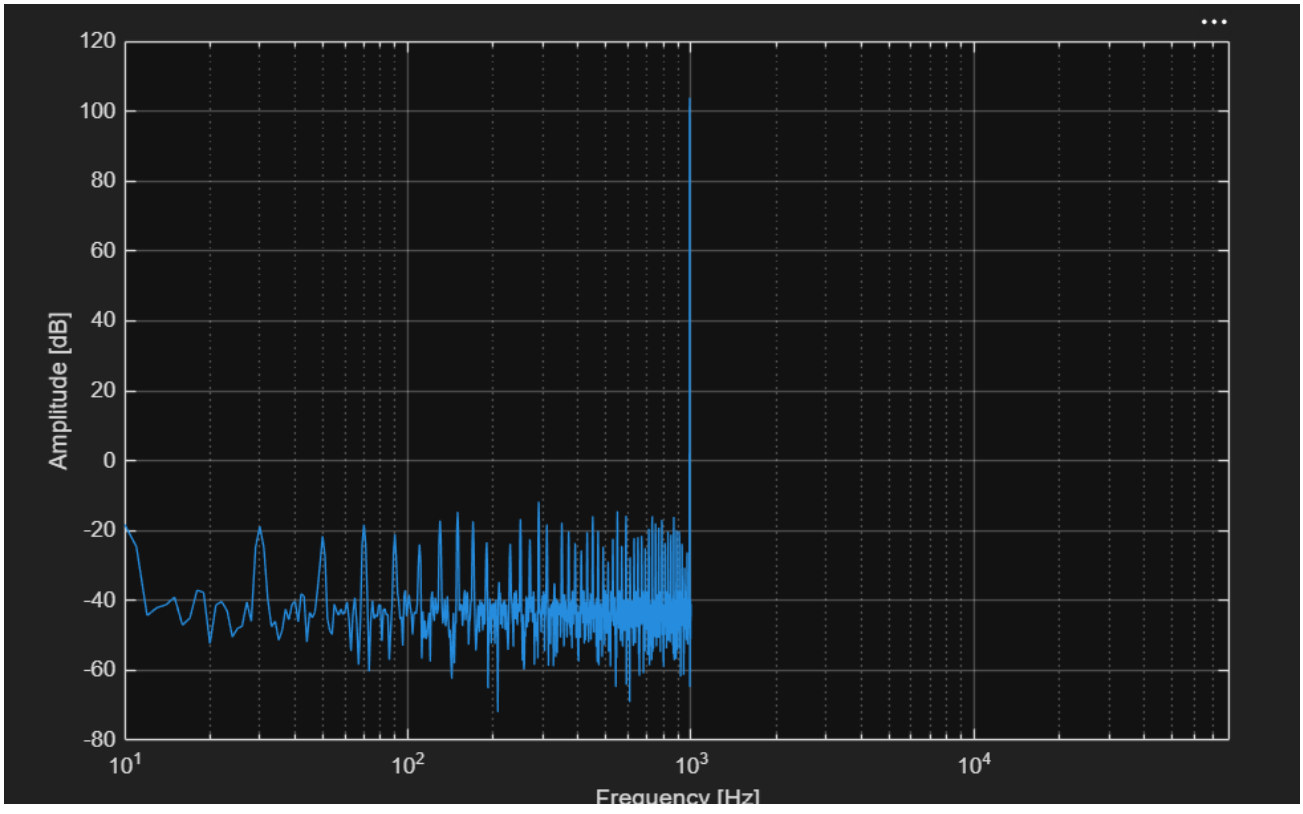
1.1. 10Hz



1.2 500HZ

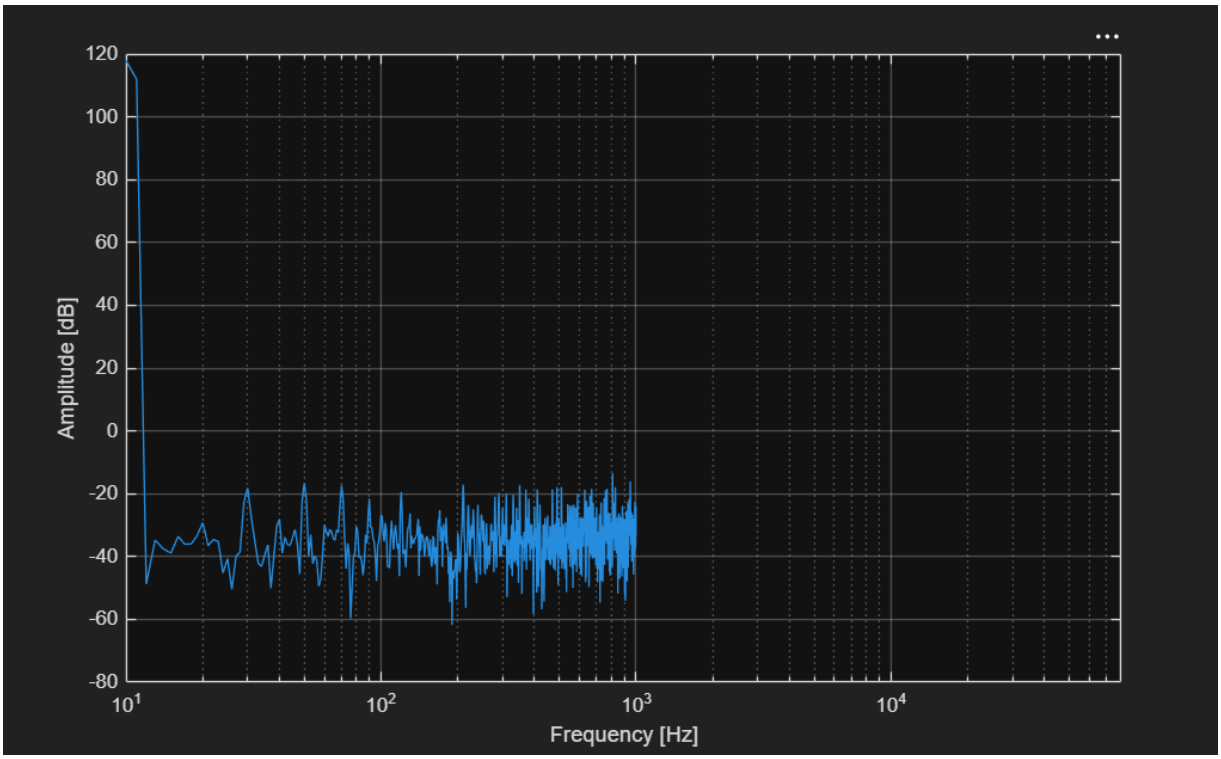


1.3 990Hz

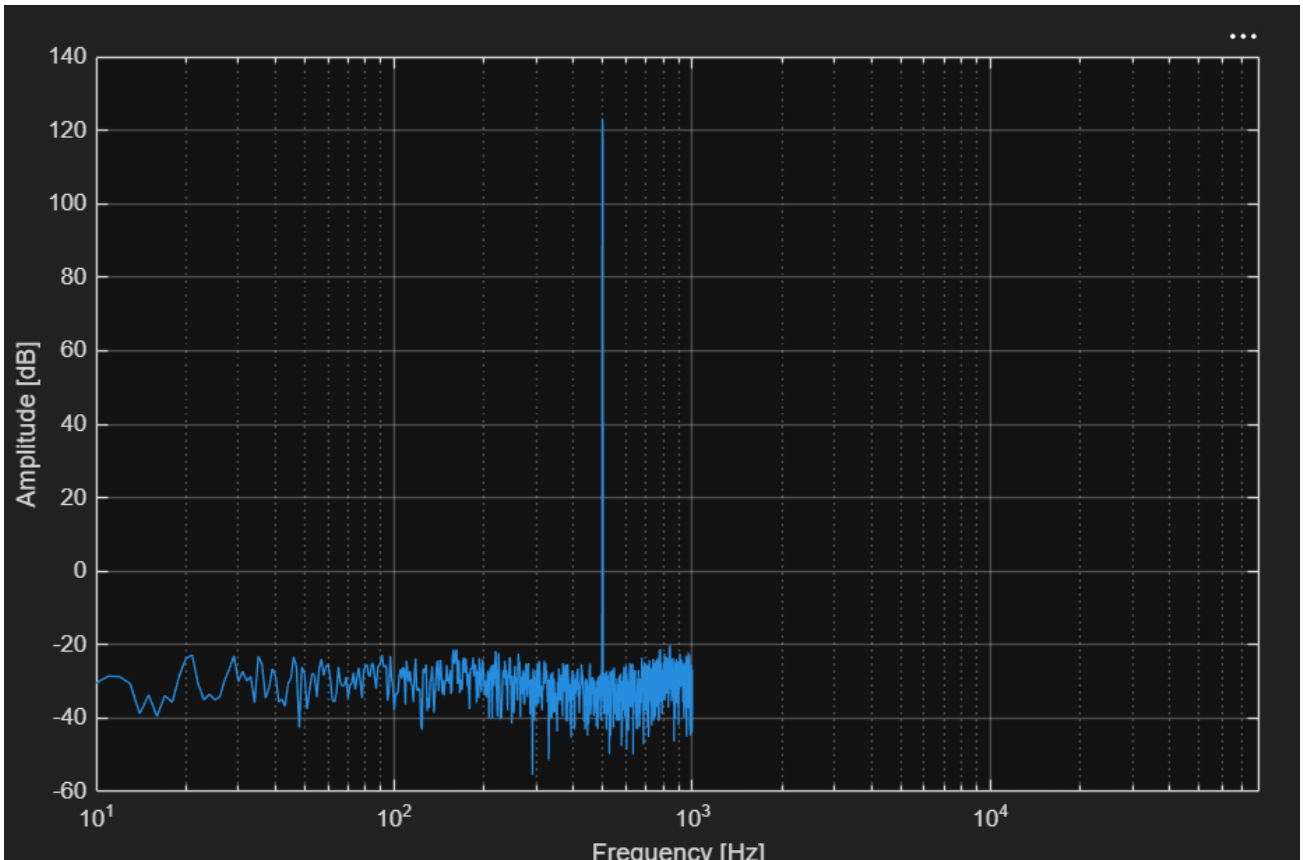


TASK 2

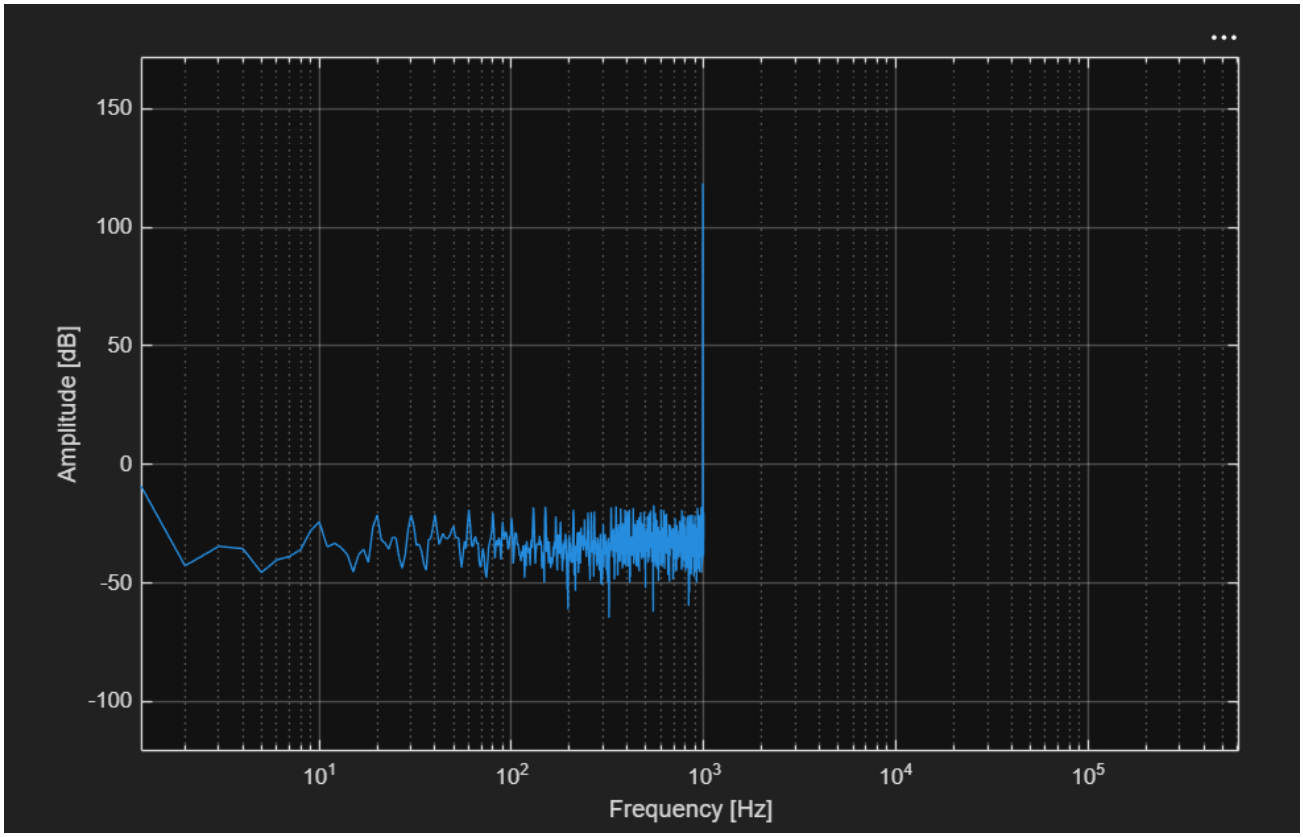
1. 10HZ



2.500Hz



3.990HZ


150
100
50
0
-50
-100
Amplitude [dB]
10^1
10^2
10^3
10^4
10^5
Frequency [Hz]